

Stakeholder Identification in Agile Software Product Development Organizations

A model for understanding who and what really counts

Ken Power
Cisco Systems, Inc
Galway, Ireland
ken.power@gmail.com

Abstract— Stakeholder Theory is an area of strategic management that defines a stakeholder as someone who affects or is affected by the actions of the organization. The principles and concepts of stakeholder theory can be applied to software development organizations to give managers a better understanding of the diverse community of stakeholders that influence product development efforts. A deeper understanding of the broad community of stakeholders can help an organization that is transitioning to agile methods by highlighting the stakeholders that are affected. This paper describes the application of stakeholder theory to product development organizations in the context of a global product development company undergoing a multi-year transition to agile development. A model is presented for mapping stakeholders into stakeholder groups, and for quantifying the influence of stakeholders. This paper further uses a stakeholder approach to demonstrate how traditional organization roles map to roles in an agile product development organization.

Keywords—stakeholder; stakeholder group; salience; agile; lean; product development

I. INTRODUCTION

There is a body of knowledge that exists in agile [1] and lean development [2-5], organization patterns [6], and systems design that can help teams become more productive and understand better the context in which they are creating products. Separately, there is a body of knowledge in stakeholder theory that helps organizations understand the needs and influences of different stakeholder groups [7-12]. A particular focus of this research is on the application of stakeholder concepts to software development, and specifically to geographically distributed product development organizations.

II. MOTIVATION FOR RESEARCH

This research attempts to understand how managers of an organization can better identify and engage with the stakeholders that influence, and are influenced by, its product development efforts. This research develops a model for

understanding how agile and lean principles combined with stakeholder concepts apply to a product development organization. In other words, this research develops a model based on stakeholder concepts that helps managers to identify and engage with the stakeholders that influence, and are influenced by a product development effort. This same model can be applied not just to an organization of people but also to something that has value to the organization, such as a product or system architecture [13].

III. RELATED WORK

Stakeholder Theory is an area of strategic management that provides models for identifying and mapping stakeholders [8, 9]. Freeman identifies seven techniques for creating value for stakeholders, including stakeholder assessment, stakeholder behavior analysis, understanding stakeholders in more depth, assessing stakeholder strategies, developing specific strategies for stakeholders, creating new models of interaction for stakeholders, developing integrative value creation strategies [9].

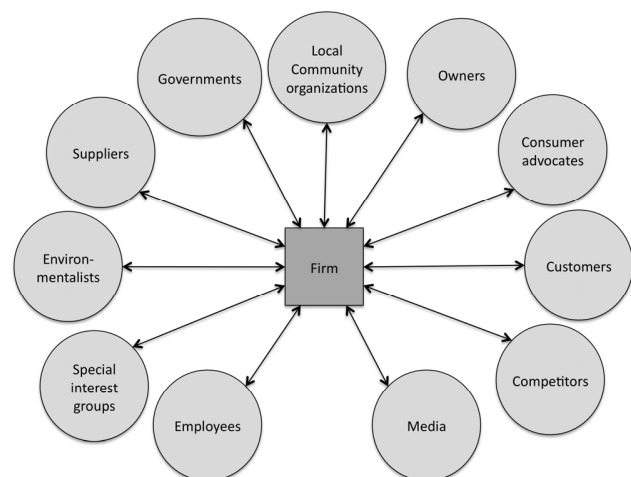


Figure 1: Typical stakeholder map of 'The Firm'

Traditional stakeholder theory places 'The Firm' or the corporation at the center and attempts to understand who are

the stakeholders in the firm. The diagram in Figure 1 shows a typical stakeholder map of the firm [9, 14].

While it is informative to understand the stakeholders in a firm, it can quickly become unwieldy to try to manage all stakeholders at once. Freeman developed a two-tier model that distinguishes between primary and secondary stakeholders [9]. The diagram in Figure 2 shows a typical stakeholder map using Freeman's two-tier stakeholder model. This map shows the firm at the center, and shows the primary and secondary stakeholder groups in the firm.

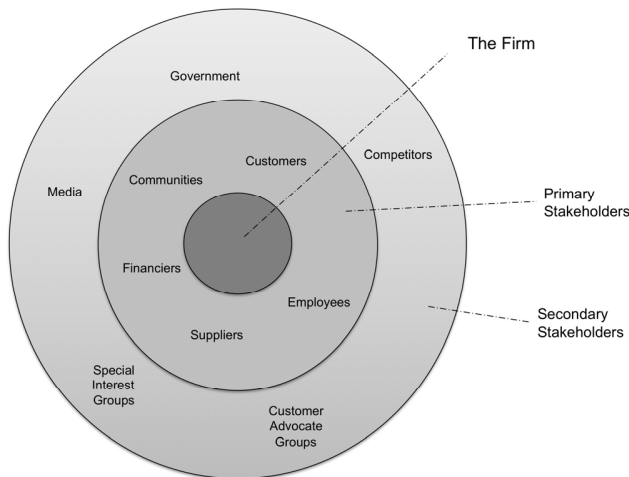


Figure 2: Two-tier stakeholder map

Some stakeholders will naturally have a higher degree of relative importance or urgency. Saliency, the principle of defining who and what really counts to an organization, is key to understanding the stakeholders that managers should pay most attention to. Mitchell, Agle and Wood define a model for classifying stakeholders based on attributes of power, legitimacy and urgency that helps managers understand and quantify the degree of salience a stakeholder possesses [15]. The category of salience of a stakeholder or stakeholder group can change over time. Therefore, salience has a dynamic aspect. The diagram in Figure 3 shows the different qualitative classes of stakeholders as described by Mitchell's model.

We can evaluate and compare the salience of different stakeholders and stakeholder groups by understanding where they fit in Mitchell's model. According to this model a stakeholder can possess none, one, two, or all three of the attributes of power, legitimacy and urgency. Stakeholders that possess none of the attributes are classified as non-stakeholders (8). Stakeholders that possess all three attributes are definitive (7) stakeholders. Latent stakeholders possess one attribute, and are dormant (1), discretionary (2) or demanding (3), depending on which attribute they possess. Expectant stakeholders possess two attributes, and are dominant (4), dangerous (5) or dependent (6).

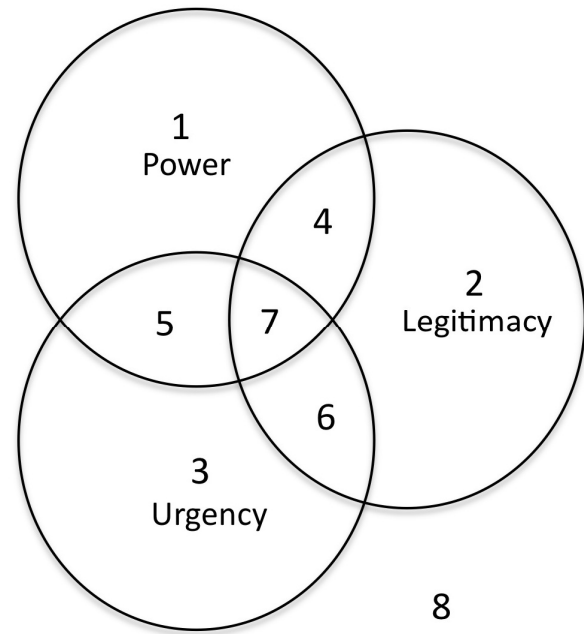


Figure 3: Qualitative classes of stakeholders

Phillips brings a deeper ethical dimension to the principles of stakeholder identification, discussing concepts of fairness when dealing with stakeholders [11]. Svendsen presents a collaborative approach to stakeholder relations that includes a business case that explains why a model of stakeholder engagement is vital to corporate success [16]. She provides a framework for understanding the importance of stakeholder relations, and, building on the work of Freeman, defines a guide to building those relations.

The organization patterns identified by Coplien and Harrison directly address specific stakeholders, and the relationships between stakeholders [6]. Viewed in this context the patterns provide guidance on building effective stakeholder relationships within software development teams. The authors consider an organization as a social system with structures, and in that context they identify patterns that have helped software companies to achieve improved efficiencies in organization and performance. They group the patterns into four pattern languages, each of which addresses a particular set of problems. There is a pattern language for project management, piecemeal growth of the product and process, organization style and role relationships, and people and code.

Although the term stakeholder is often used in software development, it is predominantly limited to customers, end-users, or project sponsors. The field of software development does not focus on product development teams as a diverse group of stakeholders with diverse interests. Where there is deeper mention of stakeholders, they typically treat the development team or 'the customer' as a single stakeholder. As far as I can tell so far, there is currently no formal use of Stakeholder Theory applied to software product development

teams, and software development processes in general. I have so far been unable to identify any application of stakeholder theory in the context of agile and lean development. A specific focus of my research is the application of stakeholder theory to product development organizations that use agile and lean processes.

IV. THE CASE STUDY

This research is applied to a product development organization that is distributed across multiple locations and time zones in the US, Europe and Asia. The organization is responsible for multiple portfolios of enterprise client and server products. The products are for general sale to volume markets, as distinct from custom products for a single customer. The majority of the products are installed by customers and integrated with their enterprise infrastructure, as distinct from hosted solutions, or Software-as-a-Service models. As with many such organizations there is no single customer or end user archetype, but rather a wide variety of customers and users that will use the products in different ways. A central premise of many agile development processes is the notion that the Customer is part of the team. For large organizations creating products for general sale, there is no single customer. This creates both a challenge and a conflict that must be reconciled for such organizations to adopt agile development processes successfully. The intent behind these Agile processes is still valid, i.e., to have the voice of the customer heard in all business-related decisions, to create a product that addresses the real needs of the customer, and to create a process that allows the customer to provide early and continuous feedback to the people creating the product, and to allow for that feedback to influence the direction of the product even as it is being developed, and to allow the team to alter course in a meaningful context if the business environment changes. The organization has adopted Scrum as its agile project management framework [17, 18], and also draws heavily from other agile processes to compliment Scrum, e.g., XP for technical practices [19]. The application of stakeholder concepts considers the different agile roles presented by these processes, and helps relate agile roles to the organization's traditional roles and job titles.

The data in this paper is drawn from a two-year period of observation, interviews and retrospectives in the organization, as the organization moved through multiple product releases while also embarking on a transition to agile development. Multiple revisions of the stakeholder groups were reviewed, leading to the model presented in this paper. The stakeholder concepts and model of stakeholder groups are being actively applied to product teams.

V. STAKEHOLDER GROUPS IN SOFTWARE PRODUCT DEVELOPMENT

There are a wide variety of stakeholders that have a stake in product development. Many teams transitioning to agile processes fail to pay due attention to all necessary stakeholders. Sometimes this is through lack of awareness. Sometimes it is through lack of opportunity, e.g., it is not

uncommon for agile processes to begin in a development group, and maybe later spread to include a test group. Even that is not sufficient. Without wider support throughout the organization the efforts of the engineering teams to become agile will be limited. Likewise, a product management or project management organization can try to apply agile concepts for portfolio and product management, or try to apply agile project management principles, but unless the engineering teams have a solid foundation in agile development and test practices the effort will fail.

VI. STAKEHOLDER MAPPING

A. Identifying the stakeholders

Many agile adoption efforts fail or become difficult because of a failure to identify and engage with all of the necessary stakeholders. Table 1 shows many of the different stakeholders in the product development organization. The table is not exhaustive, but it does illustrate the broad diversity of stakeholders in a product development organization, and the scope of the challenges faced by agile teams engaging with these stakeholders.

Software Developer	User Experience	Account Manager
QA Engineer	Interaction Designer	Technical Support
Test Automation	Universities, Colleges	Investors
Feature Test	Research Institutes	Accountant
System Test	Technical Writer	Sales Manager
Performance Test	Development Manager	Solution Architect
Alpha Test	QA Manager	Lab Administrator
Early Field Trials	Product Manager	Scrum Master
Marketing	Program Manager	Agile Coach
Enterprise Architect	System Architect	Software Architect
Security Engineer	Release Manager	Trainer
HR	Lawyer	Media
Customers	Recruiter	Support Engineer
Channel Partner	Partners	Technical Lead
Engineering Director	Distributor	Environmental Groups
Vice President	Product Management Director	Sales Engineer
Senior Vice President	Resellers	Technical Marketing Engineer
President	Integrators	Analysts
CEO	General Manager	CIO
Open Source community	CTO	Stock holders
Standards Bodies	3 rd Party Developers	Special Interest Groups
Operations Manager	Regulatory Bodies	Stock Exchange

Table 1: Stakeholders in a product development organization

B. Mapping stakeholder groups

For large-scale product development efforts we categorize the stakeholders into six primary stakeholder groups using techniques from Freeman [9]. These are Product Owner Team, Product Delivery Team, Program

Sponsor Team, Product Consumers, Product Council and Program Core Team. The diagram in Figure 4 shows the primary stakeholder groups in a product development organization.

We distinguish between product owners and customers, because they are generally different stakeholders in the context of a product development organization. The following sections describe these stakeholder groups in more detail. There is a guiding principle for identifying those stakeholders that should be part of the group. We identify relationships between the stakeholder group and current agile processes. A description of the basic responsibilities of the members of the group is provided, followed by a description of sample stakeholders from the Case Study that are part of that group.

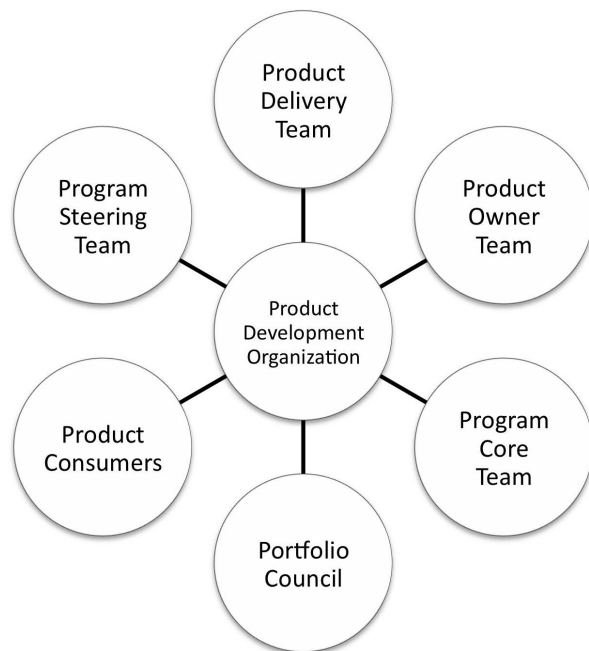


Figure 4: Primary stakeholder groups in a product development organization

C. Product Owner Team

1) Guiding principle

Anyone who has a stake in guiding the end deliverable, either through providing input to or reviewing and approving the output of, the product delivery teams.

2) Agile context

This is an extension of the Product Owner role in Scrum terms. Roth describes the concept of a product owner team as a way to fill the role of product owner on Scrum teams [20].

3) Responsibilities

We can have multiple Product Owner Teams in a large product development organization. Typically, we try to have one Product Owner Team per product.

The Product Owner Team engages directly with stakeholders from the Product Consumers. There are a wide variety of sources of requirements, so this team must bring product requirements to the table from all these sources. These stakeholders contribute requirements to the product and provide input, direction, feedback, and approval as the product evolves. This team manages the backlog, and works with the Delivery Team to make sure the backlog is prepared for release planning and iteration planning sessions.

This is a diverse team that ultimately must speak with one voice. In the course of gaining consensus on product direction, this group must decide priorities, debate trade-offs and make the best decision possible at the time with the information available. This group writes requirements as user stories, and defines acceptance criteria. People from this group engage with customers, both internal and external. They plan ahead to make sure teams always have what they need for release and Iteration planning, and they continually refine and communicate long-term product vision. In the course of all of this, they need to strike a balance between customer value and user value. This group resolves tensions and trade-offs between customer desires, user needs, user experience requirements, and technology constraints.

4) Stakeholders

Typical roles that are part of a product owner team can include product managers, engineering managers, architects, engineers, user experience designers, testers, and sales engineers. In the Case Study, product managers lead product owner teams. However, these teams must have broad representation from other functions including development, QA, user experience, architecture, and sales. User experience research is an ongoing activity in the organization, and is separate from an individual project. This is why it is important that user experience be represented as a stakeholder in this group for those products that have a significant user experience element.

D. Portfolio Council

1) Guiding principle

These are stakeholders that are responsible for multiple products that together form a portfolio. The Portfolio Council can be responsible for multiple portfolios of products in the organization.

2) Agile context

The Portfolio Council is a means of filling the Product Owner role in Scrum [20], but at a higher level. This organization uses the Portfolio Council to supplement the Product Owner Team. A portfolio council can provide direction for many Product Owner Teams.

3) Responsibilities

They interface with multiple Product Owner Teams to define the strategy, vision and direction for multiple products and/or families of products. The Portfolio Council is responsible for the roadmap that guides the product owner teams.

4) Stakeholders

Stakeholders include directors of engineering, directors of product management, senior architects, and senior engineering managers. It could also include VPs or other executives. Product lines can be managed from a product council. Hence architects and managers with responsibility for product line architectures should be part of the product council.

E. Product Delivery Team

1) Guiding principle

These stakeholders have a stake in building the product, and usually are responsible for completing tasks associated with a User Story.

2) Agile context

This is the Delivery Team in Scrum [21], and any of the XP roles that have responsibilities for completing tasks for user stories [19], e.g., testers, interaction designers, technical writers, programmers, and architects.

3) Responsibilities

These stakeholders are responsible for delivering quality products that meet the needs of the business. They perform the release planning and iteration planning activities, and are responsible for sizing and estimating their work. They commit to report their progress to each other on a daily basis.

4) Stakeholders

Stakeholders in this group form cross-functional delivery teams. These people include software developers, testers, user interface designers, technical leads, architects, and documentation writers.

F. Product Sponsor Team

1) Guiding principle

A stakeholder in this group is someone who has a stake in the product and how it fits with the strategic direction of the product development organization.

2) Agile context

XP talks about the role of executives in sponsoring agile teams, and the importance of continually ensuring that agile teams are aligned with corporate goals [19]. Crystal describes the role of executive sponsor as '*nourishing the team with decisions and money*' [22].

3) Responsibilities

These stakeholders provide executive sponsorship, help remove barriers at the wider organization level, and provide feedback on product direction via Product Owner Team. Part of this team's responsibility is representing the agile product development team in a positive way to the wider organization [19].

4) Stakeholders

From this Case Study, examples of stakeholders in this group include engineering directors, product management directors, Vice Presidents, and General Managers of business units.

G. Product Consumers

We classify stakeholders in this group as a group rather than as a team, because it is a diverse group of stakeholders all of which do not necessarily even know each other. We expect other stakeholders to interact with these, and to bring them in to the product development process as needed.

1) Guiding principle

A member of this group is someone who has a stake in evaluating, buying, or using the product that we build and in determining how it fits with the strategic direction of their organization.

2) Agile context

This is analogous to 'The Customer' in XP terms.

3) Responsibilities

These stakeholders use and evaluate product. They can, if given the opportunity, provide input to and feedback on the product as it evolves. The cadence of an Agile project gives them the opportunity to provide early feedback, and to influence the product direction as it evolves. At one point we called this stakeholder group the 'Customer Group' with the intent of it being a catchall for consumers of the product. However, we found that name to be ambiguous. This was reflected through reviews of this paper. In the context of responsibility, customers do not necessarily feel a sense of responsibility toward the product development organization, and particularly, there may be no relationship at all with the Product Delivery Team. However, customers do have a responsibility to their own organization, and their stake in the product is primarily around where the product meets the needs of their organization.

4) Stakeholders

For product organizations that develop volume products for general resale to a wide variety of customers, partners, distributors and channels, the notion of a single customer does not exist. Therefore we recognize all the stakeholders that can be considered as 'consumers' under this group. We further refine this group into internal customers and external customers. Internal customers are part of the wider organization or firm, but not necessarily part of the product development organization. It is also useful to further refine the distinction between customers and users. For example, a CTO in a customer organization has a different stake in purchasing a corporate instant messaging product than someone in the CTO's firm who will use that same product day to day. The CTO will be concerned about how a new messaging product will interoperate with the other enterprise products in her firm, and care about things like scalability, failover, security, support agreements, serviceability and product roadmaps. An end user in that same firm will care about the usability of the product, how it facilitates them getting their work done, and how it works with other products they use. From this Case Study, examples of stakeholders in the Internal Customers group include pre-alpha, alpha, solution interoperability testing, early field trial engineers, sales, marketing, technical marketing engineers, field support, and end users from the same company. Examples of External Customers include distribution channels, resale partners, governments, other firms, beta

customers, early field trial customers, and end users from other companies.

H. Program Core Team

1) Guiding principle

A stakeholder in this group is someone that is responsible for general program and project management activities, and overall success of the product release. In large product development teams there are project management responsibilities that go beyond the immediate delivery teams.

2) Agile context

Scrum does not recognize the roll of project manger or program manager, and leans more towards self-organizing teams with the Scrum Master role acting as facilitator and coach, and embodying many of the essential project management functions [21]. XP talks about the role of project manger, and acknowledges the contribution of the role [19]. Crystal also recognizes the role of project manager [22].

3) Responsibilities

Stakeholders in this group are responsible for and own the overall planning and scheduling at a program level. This often involves coordination between multiple delivery teams, and even multiple products. These people remove barriers at the program level, and ensure smooth running of the program as a whole. They manage all the work related to a release outside of the core delivery teams, including interfacing with the wider organization to help the products transition from ‘done’ to customer shipment. They report progress to Sponsor Teams and other stakeholders. They are responsible for presenting meaningful metrics to the wider organization. For large programs, these stakeholders facilitate and manage the Scrum of Scrums [23].

4) Stakeholders

Typical stakeholders include program managers and project managers, but can also include engineering managers.

VII. RELATIONSHIPS BETWEEN STAKEHOLDER GROUPS

The stakeholders groups that are usually internal to the product organization include the Product Delivery Team(s), Program Core Team, Product Owner Team(s) and Product Council.

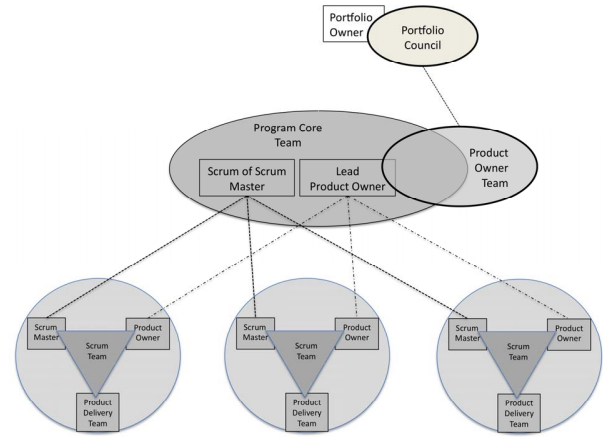


Figure 5: Stakeholder groups and agile roles

A. Filling agile roles with traditional roles

The diagram in Figure 6 summarizes how this organization chooses to fill the roles.

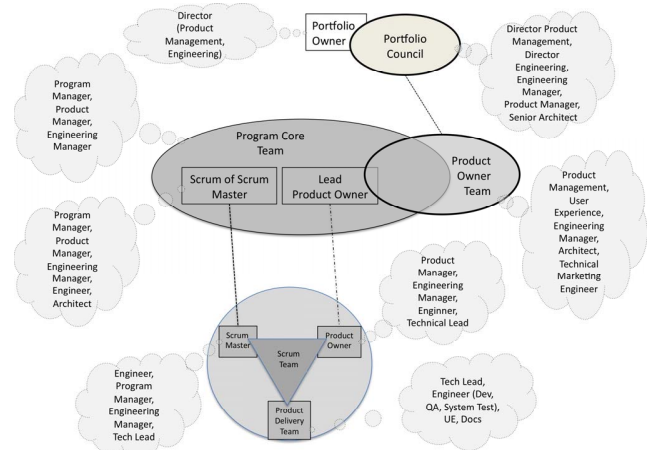


Figure 6: Mapping stakeholder groups and agile roles to traditional roles in the organization

The Scrum Master role is normally filled by an engineering team lead, but can also be filled by a program manager, engineering manager or technical lead.

A Product Manager normally fills the Product Owner role. However, in some cases the organization might not have an available product manager, or the product manager might be in a different location than the rest of the Scrum Team. In these cases an engineering manager, technical lead, engineer, or user experience designer fills the Product Owner role in a proxy capacity, and finds an appropriate time to work with the product manager. In some cases it makes sense for an engineering manager to be the Product Owner, for example if the product is an internal product, component or framework that is not itself customer facing, but gets

consumed by other products that are developed by the organization.

The Product Delivery Team is structured as described earlier under the Stakeholder Mapping section.

A Program Manager generally staffs the role of Scrum of Scrum Master, and coordinates the work of multiple Product Delivery Teams. The person in this role is responsible for interactions with external stakeholders. They focus on the multi-release product roadmap, and make sure that all stakeholders are aligned with the release plans. They facilitate the Scrum of Scrums meeting, and are responsible for removing or escalating blocking issues that are identified at the Scrum of Scrums. This person is the escalation point for individual Scrum Masters and Product Owners that belong to multiple Product Delivery Teams.

The Lead Product Owner is normally a senior Product Manager, but is also sometimes an engineering manager. This person is also a member of the Portfolio Council and is responsible for consistency of vision between the Portfolio Council and Product Owner Team.

The Product Owner Team is structured as described earlier in Product Owner Team under the Stakeholder Mapping section.

The Program Core Team is structured as described earlier in ‘Program Core Team’ under the section Stakeholder Mapping. As can be seen from Figure 6 there is some overlap between the Program Core Team and the Product Owner Team. Some people are part of both stakeholder groups, but play different roles at different times. The key differentiation is in terms of the area and scope responsibility. The Product Owner Team focuses more on the *product* on its content, while the Program Core Team focuses more on the *project* and its delivery.

The Portfolio Council is structured as described earlier in this paper. In the context of this case study the Portfolio Council is lead by a Director of Program Management who has responsibility for multiple product portfolios. Membership includes an Engineering Director, a number of senior product managers, a number of senior engineering managers, and a senior architect, all with portfolio responsibility.

B. Distinction between Portfolio Council and Product Owner Team

The primary distinctions between a Product Owner Team and Portfolio Council are that the Portfolio Council will provide high-level guidance to multiple Product Owner Teams. Product Owner Teams work directly with Product Delivery Teams, and must know the details of the product to the level where they can write user stories and acceptance criteria. The Portfolio Council has broad responsibility across multiple products. The Product Owner Team has deep responsibility within a single product.

VIII. FROM PRODUCT ORGANIZATION TO PRODUCT ARCHITECTURE

So far we have focused on the application of stakeholder concepts to the product development organization. We can apply the same concepts to a product’s architecture to help us understand the stakeholders that influence, and are influenced by, the product architecture. The stakeholder map in Figure 7 shows the stakeholders in the architecture of one of the products developed by the organization in this case study.

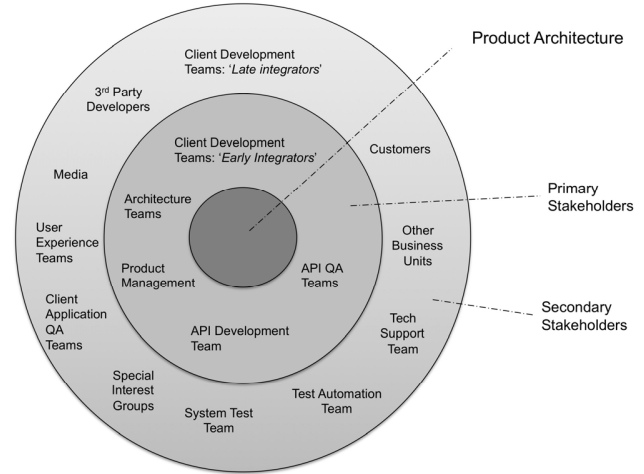


Figure 7: Stakeholders in a Product’s Architecture

This product is a middleware framework that is consumed by other applications. Identifying the stakeholders in a product’s architecture is the first step in successful engagement with them. Applying Freeman’s two-tier model, we identify primary and secondary stakeholders. Primary stakeholders typically have a more immediate stake in the architecture. This specific example is explored further in [13]. The role of architect as a key stakeholder in product development is expanded in [24].

IX. STATIC STRUCTURES AND DYNAMIC STRUCTURES

This paper has focused primarily on the static structures of the organization, and specifically on the identification of stakeholder groups, and the mapping of traditional roles to agile roles using stakeholder concepts. The dynamic structures of an organization are also important, and my research focuses on that area too. That is the subject of another paper and ongoing research [25, 26]. That work focuses on the interactions between the different stakeholder groups, and how agile processes provide ongoing opportunities for stakeholder engagement. Agile practices such as iteration planning, iteration reviews, daily stand-ups, information radiators, Scrum of Scrums, and retrospectives facilitate early and continuous identification of stakeholders, help to bring stakeholders into the development process, and provide an opportunity to address their needs. Agile practices such as continuous integration, unit testing, test driven development, and frequent releases provide ongoing feedback about the health of the product and whether it is

meeting stakeholder goals. This research also goes deeper into the implications of the salience model described earlier and examines how stakeholders transition from one category to another, and what actions an organization can take to cause or prevent such transitions.

X. FUTURE RESEARCH AND NEXT STEPS

The six core stakeholder groups are being continually reviewed and refined through engagement with multiple product development organizations. This research will develop a model that helps managers understand the broad range of stakeholders that influence, and are influenced by, its activities. This model can be applied to the firm itself, or to units within the firm, such as an individual business unit or project team. More fundamentally, this model can be applied to a software product development effort to help managers understand who and what really counts when developing products. The model includes strategies for identifying and engaging with stakeholders, and for classifying stakeholders by mapping them to one of six stakeholder groups. The research will further show, in the specific context of how agile and lean processes enable and facilitate stakeholder engagement. The model further includes a means of classifying the salience of each stakeholder or stakeholder group, and for understanding how and when stakeholders can transition from one salience category to another. This research relates how agile processes provide a means to control this salience transition and to prevent or affect it.

CONCLUSIONS

Like a software system, an organization has a static structure and a dynamic structure. This paper described how the application of stakeholder theory can help the managers of an organization to better understand the static structure in terms of the different stakeholders that influence, or are influenced by, the product development organization. It also explains how traditional roles in an organization map to agile roles.

REFERENCES

- [1] A. Cockburn, *Agile software development : the cooperative game*, 2nd ed., Upper Saddle River, NJ: Addison-Wesley, 2007.
- [2] J. K. Liker, *The Toyota way : 14 management principles from the world's greatest manufacturer*, New York: McGraw-Hill, 2004.
- [3] J. K. Liker, M. Hoseus, and Center for Quality People and Organizations., *Toyota culture : the heart and soul of the Toyota way*, New York: McGraw-Hill, 2008.
- [4] M. Poppendieck, and T. D. Poppendieck, *Implementing lean software development : from concept to cash*, London: Addison-Wesley, 2007.
- [5] J. P. Womack, and D. T. Jones, *Lean thinking : banish waste and create wealth in your corporation*, 1st Free Press ed., New York: Free Press, 2003.
- [6] J. O. Coplien, and N. Harrison, *Organizational patterns of agile software development*, Upper Saddle River, NJ: Pearson Prentice Hall, 2004.
- [7] B. R. Agle, R. K. Mitchell, and J. A. Sonnenfeld, "Who Matters to CEOs? An investigation of stakeholder attributes and salience, corporate performance, and CEO values," *Academy of Management Review*, no. 42, October, 1999.
- [8] R. E. Freeman, *Strategic management : a stakeholder approach*, Boston, Mass.: Pitman, 1984.
- [9] R. E. Freeman, J. S. Harrison, and A. C. Wicks, *Managing for stakeholders : survival, reputation, and success*, New Haven, Conn. ; London: Yale University Press, 2007.
- [10] R. E. Freeman, and J. McVea, "A Stakeholder Approach to Strategic Management," *The Blackwell handbook of strategic management*, 1999.
- [11] R. Phillips, *Stakeholder theory and organizational ethics*, 1st ed., San Francisco: Berrett-Koehler, 2003.
- [12] K. Power, "Software Development: A Stakeholder Approach," in The 11th International Conference on Agile Software Development XP2010, Trondheim, Norway, 2010.
- [13] K. Power, "Stakeholder Engagement in the Evolutionary Design of an API," in The 11th International Conference on Agile Software Development, Trondheim, Norway, 2010.
- [14] R. E. Freeman, *Strategic management : a stakeholder approach*, Boston [Mass.] ; London: Pitman, 1984.
- [15] R. K. Mitchell, B. R. Agle, and D. J. Wood, "Toward a Theory of Stakeholder Identification and Salience: Defining the principle of who and what really counts," *Academy of Management Review*, vol. 22, no. 4, pp. 853-886, October, 1997.
- [16] A. Svendsen, *The stakeholder strategy : profiting from collaborative business relationships*, San Francisco: Berrett-Koehler Publishers, 1998.
- [17] K. Schwaber, *Agile project management with Scrum*, Redmond, Wash.: Microsoft Press, 2004.
- [18] K. Schwaber, *The enterprise and Scrum*, Redmond, WA: Microsoft Press, 2007.
- [19] K. Beck, and C. Andres, *Extreme programming explained : embrace change*, 2nd ed. ed., Boston, MA: Addison-Wesley, 2005.
- [20] R. Roth, "It Takes a Village: 6 ways to fill the Product Owner Role," *Better Software*, pp. 24-28, September/October, 2009.
- [21] K. Schwaber, and M. Beedle, *Agile software development with Scrum*, Upper Saddle River, NJ: Prentice Hall, 2002.
- [22] A. Cockburn, *Crystal clear : a human-powered methodology for small teams*, Boston: Addison-Wesley, 2005.
- [23] M. Cohn. "Advice on Conducting the Scrum of Scrums Meeting," May 2009, 2008; <http://www.scrumalliance.org/articles/46-advice-on-conducting-the-scrum-of-scrums-meeting>.
- [24] K. Power, "Examining the role of architect as a key stakeholder in agile and lean product and system development," in 4th European Conference on Software Architecture (ECSA), Copenhagen, Denmark, 2010 - Submitted.
- [25] K. Power, "Software Development: A Stakeholder Approach (PhD Thesis)," J.E. Cairnes School of Business & Economics, National University of Ireland, Galway, Galway, In progress.
- [26] K. Power, "Dynamic Stakeholder Relationships in Software Development Organizations," *In progress*, 2010.